Symmetries in Physics — Exercise Sheet 6

Molecular vibrations

Variant: Questions only

Exercise 1 Show that a proper rotation through angle θ acting on $\mathbb{R}^3$ has character $\chi_{3D}(R(\theta)) = 1 + 2\cos\theta$, and that an improper operation $S(\theta)$ (rotation through θ followed by reflection in the perpendicular plane) has $\chi_{3D}(S(\theta)) = -1 + 2\cos\theta$. Specialise to the identity, a mirror plane, and the inversion.

Exercise 2 A molecule has N atoms; its displacement space is $\mathbb{R}^{3N}$, one vector per atom. Show that in the basis indexed by (atom, Cartesian component) each symmetry g acts by a permutation of 3×3 blocks, every non-zero block a copy of $R(g)$, and deduce the master formula

$$\chi_{\text{disp}}(g) = N_{\text{fix}}(g)\, \chi_{3D}(g),$$

where $N_{\text{fix}}(g)$ is the number of atoms left in place by g.

Exercise 3 A linear triatomic molecule X–Y–X (masses m, M, m, neighbouring springs k) moves along its axis only. Use the exchange symmetry of the two end atoms to block-diagonalise the equations of motion, and find all three frequencies.

Exercise 4 (mutual exclusion rule) Let the point group of a molecule contain the inversion i. (i) Show that in every irreducible representation $\Pi(i) = \pm\mathbb{1}$, so each irreducible has a definite parity (g, *gerade*, for $+\mathbb{1}$; u, *ungerade*, for $-\mathbb{1}$). (ii) Show the dipole components (x, y, z) are u and the quadratic forms $(x^2, xy, \dots)$ are g. (iii) Conclude that no vibrational mode of a centrosymmetric molecule is both infrared and Raman active.

Exercise 5 The vibrational representation of ammonia (C_{3v}) was found in the lecture to be $2A_1 \oplus 2E$. Given that z transforms as A_1 and (x, y) as E, and that the quadratics decompose as $x^2 + y^2$, $z^2 \sim A_1$ and $(x^2 - y^2, xy)$, $(xz, yz) \sim E$, determine the infrared and Raman activity of all four fundamentals. Why does the mutual exclusion rule not apply?

Exercise 6 Let the dynamical matrix D of a molecule commute with a representation Π of its point group, and suppose the vibrational representation contains the two-dimensional irreducible E with multiplicity μ. Show that the E-frequencies have a symmetry-enforced doublet factor. State the total degeneracy when an eigenvalue of the reduced matrix has multiplicity q.

Exercise 7 (boron trifluoride, end to end) Run the full vibrational analysis on planar BF_3 (point group D_{3h}: a central B with three F at the corners of an equilateral triangle). (i) Write down χ_{disp} from the master formula of Exercise 2. (ii) Reduce it, subtract translations and rotations, and obtain Γ_{vib}. (iii) Classify each mode as infrared- and/or Raman-active, and say why the mutual exclusion rule (Exercise 4) does not apply. *Data.* D_{3h} has classes E, $2C_3$, $3C_2$, σ_h, $2S_3$, $3\sigma_v$ (sizes $1, 2, 3, 1, 2, 3$; here σ_h is the molecular plane and the *operation* S_3 — not the group — is the rotoreflection: rotation by $2\pi/3$ followed by reflection in

σ_h) and the complete character table

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
A_1'	1	1	1	1	1	1
A_2'	1	1	−1	1	1	−1
E'	2	−1	0	2	−1	0
A_1''	1	1	1	−1	−1	−1
A_2''	1	1	−1	−1	−1	1
E''	2	−1	0	−2	1	0

(the class sizes are the coefficients in the headings). Translations transform as $z \sim A_2''$, $(x, y) \sim E'$; rotations as $R_z \sim A_2'$, $(R_x, R_y) \sim E''$; and the quadratics as $(x^2 + y^2, z^2) \sim A_1'$, $(x^2 - y^2, xy) \sim E'$, $(xz, yz) \sim E''$.